

Prof. Karl Friston ~ Applied Active Inference Symposium pt. 3 (Tools)

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Welcome back, everyone. This is our third session of the Applied Active Inference Symposium with Professor Carl Friston, hosted by the Active Inference Lab, and it's June 21st, 2021.

We're here representing the .tools organizational unit of the lab, the third organizational unit in the lab, and the goals of .tools is to enable effective tool and instrument use for all Active Inference Lab processes, so that's just using the digital tools affordances that we have better, as well as exploring and designing affordances for our niche, modifying our niche, resulting in effective action as well as innovative innovations in tool development. As with the other groups, we've been meeting weekly in tools and having a lot of awesome insights related to where Active Inference might come into play, and that's what we're excited to talk to you about.

Some of the core insights from the work in this unit relates to learning by doing, the recognition that modern systems are cyber-physical, everything is really intercalated with the digital, and also we found it really refreshing, kind of like a two-stroke engine, to be sidestepping or complementing or augmenting some of these philosophical discussions with technical clarifications.

And two ways in which we've seen that play out on the left here is a quote from you during a 2019 Dropbox blog post, when you wrote that technology is the natural extension of Active Inference beyond the single person,

which of course brings technology far from being something artificial into the realm of extended and embedded cognition in our niche.

And then on the right side is a slide from a very recent talk by Bert DeVry on Beyond Deep Learning, Natural AI Systems, speaking to several applications in hardware and software of Active Inference, for example, gesture recognition, robotic navigation, and also audiometry for hearing aids.

And one effort that we're starting up now is a net hack challenge.

It's kind of a video game played in text characters, and we're assembling a team with already multiple interested participants to get an Active Inference agent on the playing field, so to speak,

and have people maybe update their generative model when they see that it doesn't have to be a 3 billion parameter neural network trained for six months on the GPU,

but what if it's enough to just be curious and to want to succeed?

Those are the kinds of things that motivate us in .tools.

So we can start right off the bat with asking,

how can we use Active Inference to structure the process of innovation and tool development?

And how can Active Inference concepts help us design for complex agents that are interacting in complex niches?

For example, thinking about niche modification, extension of affordances, reduction of uncertainty, or structuring of communications?

Again, great questions.

So the use of Active Inference to structure the process of innovation and tool development.

That is in itself an entertaining notion.

In the sense that you are a realization of Active Inference, and I'm mindful that your nice use of the combination of,

you know, well, the emphasis on curiosity as the imperative that drives most of our behavior, is exactly the imperative that as a scientist drives me and most of the people I know.

And in a sense, I would imagine also drives your initiative and your laboratory.

So all the questions you are asking are really,

how do I make the next move in order to resolve uncertainty about your particular model of how, say, artificial intelligence, or human intelligence,

or human communication is going to evolve?

So, yeah, in that light, I think there are two levels to the answer.

The first one is just to celebrate and, you know, acknowledge that you are engaging in the scientific process as formulated by Active Inference,

that you are on a journey of trying to satisfy curiosity that will be never-ending.

And that speaks to your, one of your themes in the previous slide about learning is doing.

The only way you're going to resolve or sate that curiosity is to go out there and see what happens.

And that is, you know, that is exactly the right thing to do.

A more practical level answer, though, speaks to the tool development.

Because one of the, one of the fundamentals of Active Inference is the appreciation that if you just want to maximize the likelihood that this,

your kind of world model or generative model that entails the way that you exchange with and interrogate and ping a world,

is the right world that is articulated out there in terms of the sense of extended cognition, for example, in terms of the software tools or the educational tools that you're making available,

then all of this is still subject to the imperative to minimize complexity.

So in maximizing the likelihood that these tools will be out there.

And in a sense, you're saying this model or this way of narrating the way that the world works, provides an accurate description that is as simple as possible.

So there's a, you know, you cannot escape the complexity.

I'm speaking like Juergen Schmidhuber now, which is a good thing in this instance.

So that means you've got to find the simplest tools.

And it's interesting that you, you, you highlighted Bert De Vries's contribution,

because, you know, again, just practically thinking, what's the game here?

The game here is to write down the, or find the best hypothesis, the best explanation for my lived world.

And my, me could be active inference labs.

And the lived world is everything that you have to engage with in terms of educational or commercial or academic partners.

And so you've got to write down, you've got to explore the model space in terms of, to find the right model, generative model of the way that your, your system or your organization works.

And that, the first steps in writing down the generative model are basically to define its structure in terms of

the sort of hidden factors,

the latent factors and their interactions and all that good stuff.

And, but it has to be done in the simplest way possible.

And so what's the simplest way of writing down a, a generative model?

Well, it's to write down a, um, Bayesian, um, um, um, graphical model.

Um, what does that mean for the actual coding, uh, practically and the software schemes and implementation that you would either offer to people or pre-package in terms of, um, user interfaces.

Um, that it's going to be message passing on those graphs.

Um, and that, and I'm, and I'm trying to get back to Bert DeVese's, um, Factor lab, uh, Fawney, uh, Fawney lab, uh, formation.

So to my mind, that's the simplest, most generic, um, bit of computer science that you would come across in the service of finding the right software tools, um, to build absolutely everything.

Because absolutely everything can be written down as a generative model.

If there's a generative model there, there's a Bayesian dependency graph.

If there's a Bayesian dependency graph, you know, there's a factor graph.

If there's a factor graph, then, you know, there's a message passing scheme.

What is that message passing scheme?

It's just a free energy, variational free energy, minimizing message passing scheme.

So, um, I would, I would imagine that as tool development increases, that there will be a move towards a common language that would be a,

common language that will look very much like, um, Bert's, um, um, Fawney style, um, message passing.

And within that, um, you know, you've, you've got very limited choices, um, which is a good thing because that,

again, speaks to this minimization of complexity and just course graining the world, um, and your world, um, at the,

at the, at his coarsest level that will sustain an accurate account or a precise account of what you want to, want to achieve.

Um, so I'm thinking here the tools just have to come in two flavors.

They have to deal with continuous state space, um, generative models, uh, to interface with the, you know, the, the kind you need for robotics.

Um, but also, uh, the other flavor would be discrete space space, you know, latent states, um, latent, um, discrete late states, uh, models, uh, that you'll need to do say competition linguistics or, uh, you know, modeling, um, the climate in, in, in the environment.

climate in various states. And we know that the, we know all the message passing schemes that would be entailed by a commitment to one of those two kinds of models in the sense of sort of generalized Bayesian filtering for the continuous state space and by generalized, you know, I include generalized courts of motion, which generalize things like common filtering. And on the discrete state space side, you're talking about either belief propagation or variational message passing.

So when you just think about it, what you have to do in providing tools of a software kind or simulation

kind, you know, happily, there aren't many choices you have to worry about. So, you know, in that sense, all you need to do is to make sure that your tools accommodate both generalized Bayesian filtering and belief propagation and or variational message passing. And then you're using off the shelf technology, which brings us back to, well, what's the real problem then? Well, the real problem is writing down the generating model. What sort of problems, how would you unpack those problems in terms of innovation and tool development? Well, it's solving the model selection problem. So sometimes I think you said when describing the space of problems that are faced, say, with generalized AI or AGI, you know, you can unpack them at different spatial temporal scales into the inference problem, into the learning problem, and into the selection problem, by which I mean using Bayesian model selection to get the

right structure. You know, do I use six or 12 layers in my deep neural network? Do I use a convolutional model?

Or do you use a transformer? You know, these are basically problems of, that are solved if you have a mechanics

that can score the structure, enabling you to select the right form. So that, I think, is going to be a focus of innovation in the,

in the, already is, but certainly in the near future in terms of development, and in the sense that I think the inference and learning problems,

they're solved problems that you can just go to Bert and get your favorite message passing scheme, or you can keep at the level of, you know, educational or academic

message passing and use the map lab schemes that we generate, you know, here in London for toy problems.

And what is not, I think, a solved problem and will require an innovative solution is the structural learning problem or the selection problem.

Exploring not the right hypothesis, but, you know, in the principled way, exploring the space of generative models you might want to bring to the table.

Exploring not the right, you know, and that, you know, that, that has many, many different issues and, you know, things that come to mind of, of course, that you could do it in, in a, in a bottom up way by trying out new hypotheses, where do you get those from, you get them from experts in the field, because effectively they are bootstrapping themselves

in on the basis of the, you know, on the basis of our prior beliefs or our, you know, knowledge about how, how something works.

You can do it in a, in a top-down approach by, um, having, um, over-parameterized, but, um, um, over-expressive models, but with very weak, imprecise parameterizations,

and then use Bayesian model reduction to solve the selection problem.

These are ways that people are thinking at the moment, but this thinking is innovative, innovative, because I don't think there are any, any clear answers.

So, um, how would you use active inference to solve the, uh, structure of learning problem?

Well, in a sense, um, it's already being used in the sense of Bayesian model selection as natural selection, uh, but you really want to speed that up and make it work within your, within your, um, commercial academic

lifetimes.

But I would imagine the exact same principles would, would be brought to bear there.

And, you know, that almost answers the next one.

How can active inference concepts, how to design, um, complex agents interacting in complex niches.

You just have to build these things as, as, as, uh, proof of principle and hypothesis testing.

And the nice thing is, you know, all the, um, the machinery and the tools that would be requisite in building these things,

right from the, uh, variational message passing using, say, um, for any lab through to now you've got the right, um, fitness function when it comes to using, say, a genetic algorithm to explore a structure space.

Um, and what is that fitness function?

It's, it's, it's, it's the, um, you know, the evidence lower bound or the variational free energy.

Um, so you've got all the maths in place.

This is the question that I think of, of, of, of simulating these things and providing proof of principle, how you would translate that into the real world.

I, I, I, I don't know at this stage of the fact.

Um, I think, you know, a challenging first step would be, um, you know, to, to actually use, uh, robotics, uh, or in silico or, you know, sort of hardware or possibly, you know, a lot of excitement and amount of using soft, uh, soft robotics.

And actually, um, you know, in, you, you design your niche, um, and, um, see what happens and then turn your attention to, um, niche construction.

Um, where you now acknowledge that the niche itself is also succumbing to the principles, not of active inference in of, uh, in and of itself, um, in the sense that niches don't plan.

Um, but, uh, certainly, um, in the FEP sort of, um, uh, vanilla, um, free energy minimizing, uh, approach.

Um, so, yeah, I haven't actually thought about that before, but that's, uh, you know, an interesting asymmetry when it comes to simulating multi-agent interactions, uh, in the context of niche construction, where often it is the case that the, the niche is just the other agents in an ensemble.

Um, um, um, um, but if you now actually include the, the, the environment as part of the niche that is playing host to all the denizens that are the ensemble of, uh, active inference agents, um, then I, you know, I, there is this distinction between, um, the, um, um, the ability, um, um, to plan the consequences of action, um, that would entail optimization of the expected

free energy versus simply reflexively minimizing surprise or by, um, um, um, um, by minimizing, uh, free energy as a, as an evidence bound.

Um, um, um, and put that the more simply, more intuitively, um, you're either with generative models that support planning or not.

Yeah.

So there's nothing, I think, fundamentally different between these approaches.

It's just, if you've got a generative model, that is a model of the paths into the future, consequent upon what you, how you act upon the world.

That's a much richer, deeper generative model than the kinds of generative models that would be a, applicable to a thermostat or an environment.

Um, and it's likely that the environment that, you know, I have in mind here, which is, um, you know, a warehouse that you've got a sentient robot going around trying to get the right things.

So the robot can plan, but the environment, the niche can't, it'll still conform to the principles, the very, you know, the free energy principle.

There will still be particles and things that are conserved and they will still fall and behave in a predictable way.

There may even be a thermostat controlling the temperature.

None of these things are planning.

So there's a, there's an interesting asymmetry that gets into the game when you're talking about complex agents interacting in complex niches.

Part of that complexity has to be a specification of whether the complexity entails planning or not.

Um, and it just creates different problem spaces, um, certainly in the context of multi-agent, uh, simulations.

So that's, that's how I would, um, sort of carve up the, the problem spaces, um, you know, in terms of implicitly, um, problem spaces that will only be explored by doing.

And by doing, I just be, I mean, actually realizing physically these processes in the kind of situations that you think are going to be useful for, for the future.

Thanks for the answer.

And it's really fascinating about using simulation so that selection can happen within the generation of, for example, a startup rather than between generations, because of course we can let organisms or startups proliferate and then let pruning occur at the generational scale.

Or there could be ways to design so that that selection occurs within a generation, more like learning and development rather than intergenerational selection.

So awesome points there.

This could be a broad question, but we're curious what areas of applied active inference you think just might be exciting, promising or important.

So, um, my, my personal usual response to this, um, comes in, um, in two flavors.

The first is from the point of view of a theoretical biologist, um, and a psychiatrist.

So if, you know, if you can understand how a, a normal sentient artifact or person behaves, then that creates a space in which you can think about, um, false inference and false learning.

Um, or certainly, um, or certainly, um, or certainly, um, suboptimal from the point of view of, um, minimizing surprise or, or free energy.

So that's a fancy way of saying understanding the computational basis of psychopathology.

So, you know, there's a, you know, there's a whole, um, literature on using active inference as a, if you like, a normative, um, framework within which to, um, provide an ontology of false inference, um, or failures or aberrant active inference.

Um, and why would you want to do that?

Well, if it can all be reduced just to the good belief updating and the good message passing, we actually have a very, um, quite a comprehensive understanding of neuronal message passing and all its physiology and all the, um, uh, the roles of various neurotransmitters and, you know, microcircuits and your anatomy that underwrite that kind of neuronal message passing.

Um, and implicitly we also then have a fairly, um, uh, a fairly fine grained understanding of, um, the role of neurotransmitters and the pharma, uh, the consequences of pharmacological interventions in the context of, uh, experience dependent learning and, and, um, an influence of the kind we've been talking about.

So from a translational perspective, um, literally translating, um, the formalism on offer from active inference into the clinical domain, that would be one, you know, one motivation for, for, for developing this theoretical framework.

The other one is more, um, you know, uh, more in the line of technology and artificial general intelligence.

So then the question is, well, if I now want to build sentient artifacts and not only build them, but build brothers and sisters, so they are complex and interact and learn to, to love each other in a complex environment that could include me.

Um, um, you know, um, the, the, then you've got, you've got, um, a, a clear offer from active inference as to the design principles you might want to use to, to actually, um, build these artifacts.

And then there are interesting questions about, um, what the, um, what kind of artifact do you want to build?

Um, and we've already discussed the difference between a thermostat and a, uh, a sentient robot going around collecting your, um, your next sort of, um, home delivery.

Um, the, you know, there are different kinds of generative models.

So now you ask the question, okay, what are the promise, the exciting and promising kinds of artifacts, uh, as defined by their generative models, um, that one might expect to see, uh, in the future.

And then we get into the world of, um, you know, generative model support planning.

So we're talking about, um, deep generative models where they have a temporal depth.

What, what, what are the next stages that you might be looking at?

Well, there's also a sort of hierarchical depth, um, that would, um, at some point, um, first of all, include the capacity to deploy precision.

Um, and why is that important?

Well, as soon as you have, um, um, the, um, deploying the precision as a process of inference, you are now, um, you have now a normative theory or this kind of mental action or covert action.

So one example of this would be, I don't know how the, the, the technology, but I can almost, um, I can be assured that I know, I know, um, what it's trying to do.

But thinking about transformer networks and, um, the, the way that attentional selection operates in this, in this context, what you're saying is you can actually optimize the attention selection.

As an inference process using active inference or an evidence lower bound, um, where you're now predicting what things to attend to and what, what does, you know, what particular ways to switch on and which, which ways to switch off.

Um, um, um, um, and at that point you can understand that as mental action.

So when, when the transformers or variation auto encoders start to now optimize their estimates of the posterior precision, lower layers, say in an auto encoder.

It's now acquired the capacity for mental action.

And it now will pay attention to various representations and possibly even various data sources.

That's not magical.

We do that every day in the sort of MDP, uh, you know, and use it to explain a lot of, um, the attentional

mechanisms, um, implemented in the brain.

If you can migrate that technology into deep learning, you would have taken one baby step towards true sentience, which is mental action.

The next step would be, um, okay.

Um, okay.

So, um, how can I now minimize the complexity of my generative model where my generative model now actually includes this, um, meta inference in the sense I'm now providing predictions about my inference because I'm controlling the precision of hierarchically subordinate message passing.

Um, and at that point you start to think, well, perhaps one simplifying, um, one way of simplifying, uh, the computational complexity or the, you know, the complexity part of the, um, the, um, the, the inference, um, would be to carve up different states of attentional deployment in exactly the same way we're talking about carving up people into Biden versus Trump voters.

Uh, you know, a simple, uh, a simple, uh, complexity, uh, uh, complexity, minimizing carving up, which suddenly, um, suggests to you that you can now equip, um, an artifact with states of mind.

Um, so that they can be in four states of mind.

They can be happy, they can be sad, they can be confident, they can be unsure, and they will have to infer given all the evidence at hand, including the message passing there in the hierarchy, what state of mind it is in.

And if you now include in terms of the sensory evidence, you know, the voltage on their batteries or, uh, you know, some measurement of their interoception, you now have something that's getting very, very close to say Ryan's, um, notion of, um, of emotions.

So now you've got, um, a part of the generative model is now inferring what state of mind am I in as the best explanation for all these interceptive, um, embodied sensations, not just the proprioceptive, you know, state of my actuators, but also, um, you know, are they getting a bit sticky?

Uh, you know, um, are there some wear and tear, you know, are my batteries charged?

All of these things come together, um, as part evidence in conjunction with all the usual visual, you know, um, auditory, um, uh, sort of, sort of acoustic inputs, um, to, to, to, to actually, um, supply evidence for a posterior belief.

I'm in this state of mind. I'm anxious. My battery's running out.

Uh, this immediately, uh, creates different prior preferences, cost functions, if you like, that would be applied to your policies because you've got a deep generative model that comes into the future.

So now you've got an artifact that not only has the capacity for mental action, it's now got, um, the capacity to be in different emotional states.

The next step is to say, hang on. So there are these different states. Can I now equip it with a, with a minimum selfhood?

Can the hypothesis that I am actually an artifact provide empirical priors that reduce the complexity of my message passing at subordinate levels that is generating, um, that is inferring the state of mind that I am in, that in turn, um, um, um, uh, you know, optimizes the, uh,

of the precisions of various likelihood mappings or, or preferences over policies.

So at this point you're starting to get to artifacts that could have minimal self-awareness.

The next stage would be that's only going to be ever, um, um, that's only going to be, I think, useful, um,

when, when you consider dialect interactions again, because the only rationale for having self-awareness is to disambiguate, um, um, um, uh, self from other, which means that there must be some confusion.

Um, um, uh, uh, or some uncertainty, um, um, uh, at hand in order to justify the resolution and certain justify that complexity of the model.

that complexity of the model which means that you have to be interacting with or exchanging with things that are sufficiently like you to license the inclusion in your genitive model of a self versus other or that you are like me or not like me so we actually come now back full circle to what we're talking about before in terms of inferring who am i talking to so this i think there's you know structure is something quite fundamental about this inference problem are you a creature like me or not or are you like one of those are you a pet are you a plant um you know just being able to carve up this world in a way with his self-referential necessarily entails a minimal selfhood in the inferences of these that speaks to the importance of getting the necessary evidence from the environment that would um be if you like license that degree of complexity and the only kinds of environments that can supply that degree the license that degree of complexity are when that environment that you can is actually just is actually um comprises other agents like me that make it if you like worthwhile me in fact oh it's me not you doing that so i would imagine then the most promising applications um of active inference in constructing sentient artifacts pets and you know carers or or convert people things that you can converse with and will be to grow them um certainly um with themselves but but more importantly with you there so they can learn by their doing with you there so they they want to they're curious about you and you're curious about them at that point they're real one could argue that's the only scenario which you're going to have any empathetic interaction um you know you you know with these artifacts so uh i'm sure there are other applications in terms of um you know climate change or commerce or whatever but in terms of imagining what you could produce you could sell i would i would imagine that you know a mindful robot that actually is curious genuinely curious about you because that will teach it something about itself thanks for that answer um the idea of tools for attention and of design and engineering for regimes of attention to use an active inference term is really essential and what you were talking about there with the phone first off before the internet when there weren't other devices of similar kind there was no need to communicate out and what we've seen is that as there's more and more devices of similar or interoperable kinds new levels of organization have to emerge and then i thought about the anxiety that a person might feel when their phone is running low on battery right now that sensor reading is getting emotionally offloaded to the human so we could have that anxiety

on device so let's have a more relaxing relationship with our phone and then as you pointed out it would be the um incipient steps of selfhood or perhaps what they could even call a self phone if i'm allowed one pun per symposium the next question is what kinds of tools have been most helpful in your work in research which includes many areas such as spm and dcm that a lot of people who are just learning about active inference might not be very familiar with and what kinds of tools don't exist yet but might be helpful for active inference work um so the the mathematical tools so you know and i'm often asked this question

uh of students you know do i need to be able to do maths to contribute to this field um and if so what kind of maths um i won't tell you what my answer is but what i have found useful is certainly mathematics but not necessarily very high end this is always um wikipedia level mathematics and in particular um dynamical systems theory information theory and linear algebra are probably all you need to do everything really um and indeed you could read most of quantum electrodynamics as basically linear algebra with a bit of probability theory theory underneath it so that that has been the mainstay that that is you know you know if there's one tool that it would be the tool and the language um of maths and relatively simple maths and the second thing is to um the learning is doing um you know a see one do one teach one um ethos i think applies very pragmatically in this context um which means it you know it's actually um very useful if you can get students to actually build their own little um simulated artifacts um and even more useful when you know when they can actually code it up themselves which means you need access to a high level at least third generation programming language um that you know a student can get um fluent with should they want to um not only to use the existing tools but you know try and write it down themselves without having to spend years of training as a computer scientist so i found matlab very useful in that respect not because it's terribly efficient although i have to say actually some of the matrix operators and under the hood tensor operators are actually much more efficient than people give them credit for um because it actually came from um x-ray crystallography however um what's really useful about it is it uses the same syntax that you would find in a book on linear algebra um which didactically or education is really quite important when it comes to writing and reading the code um so we have deliberately stuck with matlab not because it's computationally efficient or that it's open source um it should be i don't think it is um uh but simply because it's it's configured in a way that that people reading um standard texts 101 texts and in the algebra and the like would be able to see how it transcribes into a computer language so that's been a really useful um a really useful tool and looking ahead um i imagine that you know you one's going to need open access and possibly more it's um i i um i don't know it could go out the way i'm thinking about first of all people like bert and fawny lab in terms of very generic very high-end specifications of message passing um in computer science um that may maybe that that's the level you want people to actually compose their generative models and their their artifacts and they don't even need to know about linear algebra and even less information theory what they need to know is the language of relational you know the object relations and and how to specify just different classes of exponential probability distributions and you know is it categorical is it continuous is it always positive or can it be positive and negative and that may be quite sufficient to write down a factor graph or a generative model and then everything else is just off the shelf and if you write itself um so um that would certainly be um um possibly helpful in the future and so i'm moving on to what kinds of tools dead is this at the moment so i'm thinking of it um what i never used it but i what i imagine would be that um burt's um forney lab facilities but uh offered as um an application or a user interface uh allowed you to compose generative models and then you see it compose a generative model compose a generative process the

actual world that's going to be modeled and then let's click run and see what happens that you know that would that would be that would be really useful i think um having said that the other side to to to to um future scoping here is i repeat this sort of um leveraging um more specialized or other fields uh and the you know um amortizing certain parts of of of inference or learning to infer um as a or indeed inferring to learn or learning to plan or learning to infer how you plan so or starting to sort of see what parts of the inference process are so concerned that they could actually be amortized and learned and certainly that looks as how as though that's what the brain has done for example there are people who think that the cerebellum has basically learned how the motor cortex does its online kl control or um common filtering and therefore lends a fluency and a computational efficiency to the message passing which um you know which in its absence um it doesn't mean you can't do something it just means you can't do it as fluently and as gracefully as quickly as you know as you know as you could with a cerebellum indeed when you um have a cerebellar lesion all that really happens is you become a bit clumsy um and slow um

um so um that those kinds of tools you know a quick and cheerful um integration or importing um various amortization and deep learning technology um into a um a four needs style message passing scheme that could

support you know any kind of generative model i think you know would be really really useful awesome thank you approaching this nexus from another angle what kinds of tools and platforms could inform transdisciplinary highly contextual and engaged teams that are working with these approaches act inf lab we hope to be working with others to be developing the active uh inference curriculum and body of knowledge more broadly but when teams are actually using these kinds of approaches what kinds of platforms might exist to enable their work um again i have a strong suspicion that you know the answer to this much much better so i'm trying to guess at the answer that you you know is the right answer and i'm not doing very well here um so um i you know i think that you've you you we've already talked and and certainly implicitly um in the way that you've presented the ambitions uh and implicitly send the questions and you know all the answers are there whether that's trying to engage through education whether it's trying to engage through insight um using say you know embodied experience illustrations of the basic principles whether it's supplying um um games or user interfaces graphical user interfaces to facilitate the designing and enacting and the playing with um um generative models and an active inference i think these are all um uh your obvious and laudable ways of um of you know uh of leveraging um what what what what um active inference has to have your house has to offer um um participatory yeah it's um i mean the the learning is doing thing and the see one teach one do one you know keeps coming back to mind um and and the course completely licenses the participatory aspect um but what what kind of participation did you have in mind are you talking about sort of hackathons are you talking about

some um playing games with active inference computers that that start to start to uh

um hate you or love you or or what what what what level of participation were you

yes steven do you want to give a quick thought on a few kinds of participation or what does that mean to you

yeah one area is quite interesting is in psychodrama they use action methods like action sociometry or spatial activities to look at how people relate to their experience in a dynamic way so physically so i've been looking at ways that um spatial participatory approaches can unpack people's relationships to different niches or different workplaces or different types of embodied experience and then that could be visible um to be put into active inference type geometries

i see okay right i right well there's a great example so um so two things um that i've come across before are the um architectural design um and the um the importance of um

not just uh you sort of pragmatic affordances you know can i walk up that or can i sit there but also the epistemic affordances

you know if i look over there um what would i learn about the space around me if i go around that corner so um there is um you know embryonic interest um in in you know in my world um from um this the architectural sciences and you know and architecture in and of itself that um is is is could in principle be motivated it's an odd discipline because it's half like art and half like science but

the certainly some of their ideas are very much um aligned with certainly gibsonian notions of affordance and also um the um the affordances the dual aspect affordances brought by expected free energy under active inference so it's not just you know am i the kind of creature that can sit on this on this particular chair but also what will i learn if i if i so do and so things become epistemically attractive to engage with the other domain um is um in entertainment and in music and in particular um the joy of synchronization and mutual predictability or minimizing free energy through mutual prediction when singing or dancing together or indeed um interacting with a slightly greater um asymmetry in terms of being a member of an audience um watching a band for example so um you know one of the key things that comes out of that kind of research in ways of um measuring the implicit generalized synchrony that you get from having um this information geometry that i was talking about before that rests upon there being um a synchronization manifold between the inside and the outside if the outside is another inside from another person's point of view what you now have is something called a synchronization manifold so there's a mathematical image or space um to actually talk about mutual inference and mutual active influence and engagement and communication singing together for example or um diachronically exchanging um messages uh that does actually translate mathematically into movement and belief of uh updating on a synchronization manifold and that has real world correlates you can measure that using um um kinematic um measurements you're putting leds on people who are dancing together

for example or um measuring heart rate variability or galvanic skin responses or uh doing eye tracking or indeed eeg and start to so there's quite a lot of work in um um in things like hyper scanning and in um in it you

know sort of ethology and dance uh um disciplines um where uh in the arts in the life sciences uh where they do use a lot of these techniques to quantify the degree of generalized synchrony what it would be nice to do is actually try and model that synchrony or understand that synchrony in terms of um movement on the synchronization manifold which is sort of the mutual uh belief updating and one thing which comes out of that

just in discussion if not if no if no further is the reciprocal the circular causality that is necessary
um to maintain um to maintain um that generalized synchrony um the particular synchronization manifold
we're talking about for the from the point of view of active inference of course is mediated across
the markov blanket as of the active and sensory states um but in general the the um you need to have
reciprocal coupling um um in in order to to get to get synchronization so directed coupling doesn't work
um and if that's true what that means is that engaging as an audience for example or participating
as a spectator will only really work in terms of establishing that generalized synchrony that you are
chasing and why you're chasing it well as soon as you have a generalized synchrony you've got predictability
for for free for all uh and that's a good thing because that minimizes free energy you know the
more predictable you can make the world the better the better it is um from the point of view of free
um but you can only do that if as a member of the audience um or a witness to something you can actually
actively intervene on it so that brings to mind how could you get for example a discussion discussing
this with um friends of of maxwell yeah if you wanted to promote virtual uh concerts on your online
for example during the pandemic what you don't have online which is what glues things together things
like mosh pits in in sort of carnivals and and uh festivals um is you don't have the audience participation
the the applaud the wars the you know the um the lighter waving or the light waving um so how would
you get that back into into a virtual um virtual experience because that would be absolutely essential
you know i think to actually engage people otherwise you know you you'd be you'd be looking at a pop
concert on television so you know more than just if you like um revealing the underlying correlates of that
generalised synchrony in terms of the eeg traces of the the dancers or um doing some sensory mapping from
the uh their motion to auditory import you know just making it making the sensory evidence that supports
the mutual inference um more precise and more available uh just by having it displayed say by putting
um putting motion in sound or sound in motion um or eeg electrocapographic measures of performance or
the audience

um um visualizing that and that has been done by people like paul mishore in in barcelona more than that
to actually um enable the audience to change what the performers are doing you have to make you know or
perhaps what other members of the audience are doing so you have to empower them to close the audience
to the audience to change what the performance are doing

you have to make you know or perhaps what other members of the audience are doing so you have to
empower them to close that um that circular causality to get that dynamical coupling place so you get you get
the right kind of generalised synchrony so that um you know that that sort of um dynamical systems
perspective on synchronization and free energy minimization certainly speaks to a particular kind of
participation of the audience and the audience is

that does indeed um rest upon action oriented approaches but crucially it's the action of the audience on the
performers not the performers action on the audience uh that is is is usually what you need to pay more pay
more attention

i don't know if that was that the kind of thing you were you were thinking about yeah that's really a useful
answer actually yeah we were thinking about that and and some participatory immersive theater type events
and other participation in collective meaning making so that's the type of thing that we're looking at and it

reminds me of the live stream affordance which is relatively novel but allows people to be asking questions and it enables not just efficient production of material in a one-shot approach but it allows the feedback and i can't help but add that it's that affordance for participation for example speak now or forever hold your peace that expands the wedding into the community because there is the opportunity for feedback it's not just a breakaway click it's actually something that remains integrated through the affordance for participation so i'll turn to the last question for this section

how might future modeling involve large-scale patterns in social data sets and working backwards to infer their hidden causes for example in the case of pandemic modeling governance economic other situations well this is a very practical and very prescient question because of course a lot of people are asking themselves that now is specifically with respect to pandemic models but also

the people who are exercising and have the interventional clout when it comes to covid are generally also the people who are invested in climate change problems as well so there's a lot of noise out there at the moment about

you know how we can harness the data assimilation and modeling advances made during advances made during COVID-19 and keep the momentum up to tackle climate change and not just climate but the economic structures that

and financial structures and informational structures that are deeply interwoven in terms of climate change and climate change and my answer is going to be somewhat deflationary and and i've had this kind of conversation before again with

with Maxwell and John Clippinger and related friends and i'm due actually to have another conversation with them on

um um certainly open um um open world or open i can't remember um um uh in the near future um there's a temptation to take all the high church of the free energy principle and active inference and epistemic foraging and all of that good stuff we were just talking about and say oh well now let's make it work in terms of in terms of understanding um say the pandemic and you don't need to do that all you need to do is to apply the good scientific principles that things like active inference appeal to to the problem at hand and it all comes back to the

generative model so you know all you're saying here is how might future modeling involve large-scale patterns and social data um to infer the the um hidden causes is just a statement of we need

the right generative models to make proper sense of the big data at hand and um in saying the right generative models we need the equipment both to invert those models in the sense of inferring the parameters interactions um um uh using the simple tools that we've just talked about they will just be lifting it from

from burt's laboratory or um you know continuing to use matlab um but the bigger problem is one we talked about which is the selection of the structural learning problem um so this goes beyond just um

higher you know how many layers do i have in my deep network um much more um important i think is the factorization um it's knowing how many conditionally independent factors do i need

to minimize the complexity to minimize the complexity to get the right granularity the right way of carving up um the latent causes behind all the data that is available to me um so i think the pandemic modeling is a beautiful example of this um because you know the factors that determine whether i infect you can certainly be written down in terms of virology and the um and the ace receptors and the

um and predictive validity then you can intervene on it then you can say well what would happen if i change this what would happen if i change that um and what would happen now what would happen in the future so that you know you're suddenly in a world of quantitative modeling um um where you can start to um ask some very powerful questions and also um share with everybody who matters and the products of your

inference so you can now start to think about having supplementing the weather forecast with the uh an epidemic forecast you know the virus in your area and tomorrow we expect you know the cells you can also do that to the markets and these kinds of things i think are going to be more important um when people um or when the current generation get to um your generation i guess yeah and um start to wrestle more with climate change um because they're going to want to not just know what whether it's going to rain tomorrow they're going to want to know you know at the level not just the weather but the climate what are the indicators because those indicators um really contextualize and inform their geriatric models about their place in the world and that and that global and that global scale but to provide that kind of weather forecasting that meteorology beyond the weather um you're going to need to have these you know these state space models properly optimized and you know in a first principle way in relation to the marginal likelihood of their um or their um um the the the evidence evidence bounds um just read governance here because governance is just policy decision making based upon counter-factor outcomes so that is always underwritten by um these these basing beliefs but you can't get the basing beliefs unless you've got a generative model and that has the consequences of action in the future uh there would be also um interventions either politically or financially or um or um or otherwise thank you so much again for joining this symposium it was really a special moment for the lab and we look forward to continued interaction so much appreciated and we will see you all soon thanks for everyone who's watching and we hope that you participate in act inf lab so thanks everyone bye